

AMGAD MADKOUR

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PROFILE

Engineering leader and hands-on tech lead specializing in AI-powered consumer platforms and large-scale systems. Deep expertise in LLMs, knowledge graphs, and data platforms, with a track record of building 0→1 products and scaling high-performing engineering and applied science teams. Proven ability to drive measurable growth, reliability, and efficiency improvements across systems used by millions.

EXPERIENCE

Microsoft AI, Commerce

Principal Engineering Manager

March 2022 – Present

Redmond, USA

- Built Microsoft Copilot Commerce Savings from the ground up (0→1), shipping AI-powered shopping experiences including product tracking, price alerts, coupons, and cashback to millions of users.
- Lead the addition of a savings skill to Microsoft Copilot that orchestrates product tracking, coupons, and cashback tools, and built extensive evaluation frameworks to measure skill response quality, tool-call accuracy, and reliability at scale.
- Designed and built the product tracking platform, driving 2× growth in product tracking within six months through hands-on architecture, platform improvements, and disciplined execution.
- Built Autos Copilot and My Garage from scratch (0→1), spearheading Microsoft Autos Marketplace initiatives that contributed to 3× DAU growth and 100% YoY revenue growth.
- Operate as a hands-on tech lead, staying deeply involved in system design, architecture reviews, and implementation across the full stack while leading the team.
- Built and scaled a multidisciplinary engineering and applied science team from scratch, owning hiring, role definition, leveling, promotions, and performance management.
- Partner with product, design, business, and engineering leaders to define roadmap and drive execution across teams.

Microsoft Bing

Senior Data & Applied Scientist

January 2019 – March 2022

Redmond, USA

- Designed, developed, and fine-tuned ML models for product information extraction in Microsoft Edge, improving accuracy and production reliability at scale.
- Reduced compute and storage costs for entity matching workflows by **50%** through optimized indexing and candidate generation.
- Scaled the Bing Satori knowledge graph entity matching platform on Apache Spark and pioneered adoption of .NET for Spark within the Bing stack.

IBM

Research Engineer

February 2006 – September 2010

Cairo, Egypt

- Co-invented a patented k-partite graph technique for tag recommendation during resource bookmarking.
- Developed (Bionoculars), an unsupervised biomedical information extraction system for identifying protein-protein interactions from scientific literature.
- Created a machine-assisted human translation system (MAHT) leveraging language models for statistical machine translation.
- Collaborated with researchers in IBM Watson Research lab to build an unstructured information management platform for knowledge discovery from news.

EDUCATION

Purdue University
Ph.D. in Computer Science

December 2018

PATENTS

Selecting a data element in a network, US Patent 8,589,409.

SELECTED PUBLICATIONS

WORQ: Workload-Driven RDF Query Processing. International Semantic Web Conference (ISWC), 2018.

SPARTI: Scalable RDF Data Management Using Query-Centric Semantic Partitioning. SIGMOD Workshop on Semantic Big Data (SBD@SIGMOD), 2018.

Tornado: A Distributed Spatio-Textual Stream Processing System. Proceedings of the VLDB Endowment (PVLDB), 2015.

Language Independent Transliteration Mining System Using Finite State Automata Framework. Named Entities Workshop (NEWS@ACL), 2010.

BioNoculars: Extracting Protein-Protein Interactions from Biomedical Text. BioNLP Workshop (BioNLP@ACL), 2007.

TECHNICAL SKILLS

AI/ML	LLMs, RAG, Agentic AI, LLM Evaluation, Knowledge Graphs
Technologies	Azure, AWS, Kubernetes, Databricks
Leadership	Team building, hiring, mentoring, cross-functional execution